

BIND9 DNS Assignment - Questions and Answers

Question 1: Install BIND9 DNS server on your Linux machine and verify the installation by checking the version with the `named -v` command.

Answer:

Install BIND9:

```
sudo apt update
```

```
sudo apt install bind9 bind9-utils bind9-doc -y
```

Verify Installation:

```
named -v
```

Example Output:

```
BIND 9.x.x
```

Question 2: Configure a basic DNS zone file for `myzomatoclone.local` mapping it to `127.0.0.1` and restart DNS service.

Answer:

Edit `named.conf.local`:

```
sudo nano /etc/bind/named.conf.local
```

Add:

```
zone "myzomatoclone.local" {  
    type master;  
    file "/etc/bind/db.myzomatoclone.local";  
};
```

Create Zone File:

```
sudo cp /etc/bind/db.local /etc/bind/db.myzomatoclone.local
```

```
sudo nano /etc/bind/db.myzomatoclone.local
```

Add Records:

```
@ IN A 127.0.0.1
```

Check Configuration:

```
sudo named-checkconf
```

```
sudo named-checkzone myzomatoclone.local /etc/bind/db.myzomatoclone.local
```

Restart Service:

```
sudo systemctl restart bind9
```

Question 3: Set your system's DNS resolver to use your local BIND9 server and test resolution.

Answer:

Edit Resolver:

```
sudo nano /etc/resolv.conf
```

Add:

```
nameserver 127.0.0.1
```

Test Resolution:

```
dig myzomatoclone.local
```

or

nslookup myzomatoclone.local

Expected Result:

127.0.0.1 returned as the IP address.

Question 4: Add a CNAME record for foodsearch.myzomatoclone.local and verify it works.

Answer:

Edit Zone File:

```
sudo nano /etc/bind/db.myzomatoclone.local
```

Add:

```
foodsearch IN CNAME myzomatoclone.local.
```

Reload DNS:

```
sudo systemctl restart bind9
```

Verify:

```
dig foodsearch.myzomatoclone.local
```

Expected Result:

Alias resolves to myzomatoclone.local and returns 127.0.0.1.

Question 5: Find and fix the error in this zone file snippet.

Answer:

Incorrect Snippet:

```
myzomatoclone.local. IN A 127.0.0.1
```

```
foodsearch IN CNAME myzomatoclone.local.
```

Issue:

The records must be placed on separate valid lines within the DNS zone file and the CNAME target should end with a dot when using a fully qualified domain name.

Correct Configuration:

```
@ IN A 127.0.0.1
```

```
foodsearch IN CNAME myzomatoclone.local.
```

Verification:

```
sudo named-checkzone myzomatoclone.local /etc/bind/db.myzomatoclone.local
```

```
dig foodsearch.myzomatoclone.local
```

Expected Result:

The CNAME record resolves correctly to myzomatoclone.local.